Verify Important Outputs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

TEACHER GUIDE · 1 OF 9

Verify Important Outputs

> Prepare Verify Important Outputs: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Separate an answer into **claim**s and connect each **claim** to relevant **evidence**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, **independent** cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Prompts Are Design Instructions

Without opening the old lesson, explain one key idea from Prompts Are Design Instructions. Give an example and a mistake that the idea helps you avoid.

Look for: Write a prompt with a clear task, supplied **evidence** and checkable **output** requirements

Reference example, not the only acceptable wording: Possible omissions include audience, **source** facts, required fields, format, missing-value behaviour and limits on extra **claim**s.

C supplies **evidence**, Grade 6 specifies readers, and four labelled lines define visible fields and structure. These requirements support checks but do not guarantee compliance.

Privacy by Choosing Less

Without opening the old lesson, explain one key idea from Privacy by Choosing Less. Give an example and a mistake that the idea helps you avoid.

Look for: Justify which information a stated task needs and remove unnecessary fields

Reference example, not the only acceptable wording: Keep topic choice; remove name, full date of birth, address and photograph. The totals do not require identity or contact details.

Collecting extra fields creates copies and exposure to manage even if those fields are later ignored or deleted.

Good Data, Bad Data

Without opening the old lesson, explain one key idea from Good Data, Bad Data. Give an example and a mistake that the idea helps you avoid.

Look for: Check records against a stated format, **unit** and meaning before using them

Reference example, not the only acceptable wording: Height 60 fails the whole-number range 1 through 10. Travel blank is missing; its value cannot be inferred.

Travel 0 is valid in the whole-number range 0 through 10 and can represent observed no travel. Zero is a measurement, not a replacement for a blank.

Use the student module and workbook section with this exact week title. The worked case, **independent** question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A generated report says 18 of 24 tests passed, so success was **80%**. Verify the **claim** in two ways and write a corrected sentence.

Concept to explain

Fluent AI **output** is not **evidence**; important **claim**s need **independent** support.

Term	Meaning
claim	statement that can be checked
source	origin of evidence
independent	not merely copying the same origin
corroborate	support a claim with separate evidence

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; **independent** assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Use the supplied fictional **source** cards to keep the activity accessible offline and separate **evidence** checking from internet search skill.

Supported and contradicted are judgments relative to stated **evidence** and assumptions; unresolved is not a synonym for false. Emphasise that identical repeated **claim**s may share one origin.

This lesson checks exact paper models, not uncertain physical measurements.

Readiness answer

No. Format and factual support are separate criteria. A correct layout cannot make an unsupported time acceptable.

Current ribbon card R and old card Q details in the lesson

A metre strip marked into 100 centimetres or a paper drawing of one

The **independent** craft card M in W4 and paper for **claim**-by-**claim** audit notes

End goal for the pictorial case

By the end, I can verify quantities with consistent **unit**s and separate arithmetic from unsupported **claim**s.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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Explain the mechanism

> Model Verify Important Outputs using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT specific **claim** extracted from an **output**

2. Process — the deciding step

PROCESS check **independent** suitable **source**s and reconcile differences

3. Output — the result

OUTPUT supported, corrected, uncertain, or removed **claim**

Key idea

CORE IDEA Fluent AI **output** is not **evidence**; important **claim**s need **independent** support.

How the mechanism works

UNDER THE HOOD **Verification** is **claim**-level **evidence** work. **Source** independence, authority, direct support, recency, and agreement matter. Multiple pages repeating the same upstream error are not **independent** confirmation.

Draw a fresh example with all three stages:

Every step connects a **claim** to observable **evidence**.

The case to investigate

CASE A generated paragraph confidently **claim**s that an animal lives in a region; the wording is not **evidence** that the **claim** is true.

Step 1

STEP 1 — OBSERVE specific **claim** extracted from an **output**

Step 2

STEP 2 — TRACE check **independent** suitable **source**s and reconcile differences

Step 3

STEP 3 — CONCLUDE supported, corrected, uncertain, or removed **claim**

Model answer

MODEL REASONING Check the exact **claim**, not the whole paragraph. Use **source**s with direct relevant **evidence**, note publication context, and explain whether they **independent**ly agree.

If **evidence** is weak or conflicting, revise or remove the **claim**.

A claim to avoid

DO NOT OVER**CLAIM** Two search results are not two **independent source**s when both copy the same page.

What new **evidence** could change this answer?

Concrete teaching case

The teaching case

An answer says six groups of four make 28.

The teaching case

Verify **independent**ly.

Exact explanation to model

Why the result follows

Six times four is 24. Draw six groups or add four six times. Confident wording does not

> change arithmetic.

Explain the added worked example

Explain the added worked example · 1

An answer can contain a mixture of correct, incorrect and unverified statements. Checking one sentence does not validate every other sentence. First separate the answer into individual **claim**s.

Then ask what **evidence** would support each one: a record, a measurement, a calculation or some combination.

Explain the added worked example · 2

Our fictional ribbon activity uses current materials card R. R states: four rolls of ribbon, each exactly 2 metres long; twelve pieces will each use 50 centimetres; no cutting loss is included in this paper model.

One metre equals 100 centimetres. An old card Q describes a different activity with three rolls. These are provided classroom records, not instructions to cut real materials.

Explain the added worked example · 3

A draft says “R gives four rolls. The starting length is 8 metres. The pieces use 600 metres. Two metres remain. Every participant will enjoy the activity.” Do not approve or reject the whole paragraph at once.

Inspect the roll count, starting length, used length, remaining length and enjoyment **claim** separately.

Explain the added worked example · 4

A **citation** is useful only if you can locate the **source** and confirm that it supports the specific **claim**. The title “Materials card R” beside a paragraph does not make the enjoyment **claim** part of R.

If a cited card Z is not supplied or cannot be found, that **citation** cannot currently verify its **claim**. Record the gap rather than inventing Z’s contents.

Explain the added worked example · 5

Check whether the **source** covers the right activity, version, population or time. Q's three rolls may be accurate for its own activity but irrelevant to the current four-roll task.

In real work, also inspect who created the record, how they obtained the information and whether there is a correction or better **source**.

Key terms

Term	Meaning
Claim	A statement that can be examined against evidence
Verification	Checking whether a claim is supported by appropriate evidence
Citation	A reference that helps a reader locate a source
Unit	The agreed measurement scale attached to a quantity, such as centimetres
Corroboration	Additional evidence that supports a claim, with its origin checked

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the **independent** case answer during modelling.

Quantity	Calculation in centimetres	Equivalent in metres
Starting length	4 × 200 cm = 800 cm	8 m
Length used	12 × 50 cm = 600 cm	6 m
Length remaining	800 cm − 600 cm = 200 cm	2 m

Model the reasoning aloud · 6

Convert to one length **unit** before subtracting. Two metres is 200 centimetres, so four rolls give 800 centimetres. The twelve pieces use 600 centimetres, leaving 200 centimetres, or 2 metres.

Subtracting the bare numbers 8 – 600 would mix metres with centimetres and would not describe the remaining length correctly.

Model the reasoning aloud · 7

Use a second route as a check: 50 centimetres is half a metre, so twelve pieces use six metres. Eight minus six leaves two metres.

A reasonableness check also asks whether the used length is at most the starting length and whether starting length equals used plus remaining.

These checks can reveal mistakes; they do not replace checking the original **input**s and assumptions.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Verify Important Outputs

A fictional craft sheet calculates ribbon left after making bookmarks.

The facts we will use

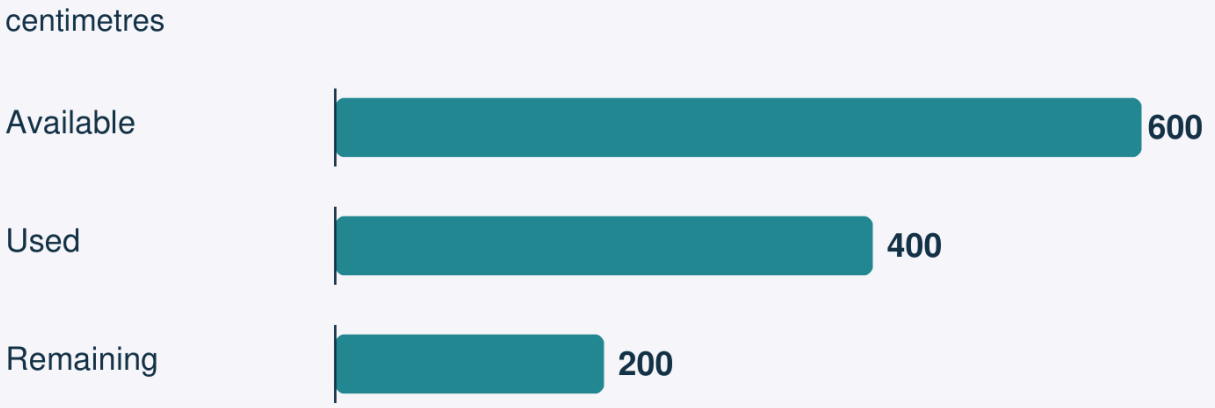
> **Supplied facts**
>
> Four rolls each contain exactly 150 cm. Eight bookmarks use exactly 50 cm each.

> **Supplied facts**

No other ribbon loss is assumed; no competition result is recorded.

AI ACADEMY / AI-1/27

Ribbon quantities in one consistent unit



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/27

Verify Important Outputs

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Available ribbon

4 x 150=600 cm.



2. Ribbon used

8 x 50=400 cm.



3. Remainder

600-400=200 cm=2 m.



4. Audit draft

"200 m remains and everyone wins" fails unit and evidence checks.

CHANGE ONE THING

Make ten bookmarks instead of eight.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Available ribbon	4×150=600 cm.
2. Ribbon used	8×50=400 cm.
3. Remainder	600−400=200 cm=2 m.
4. Audit draft	“200 m remains and everyone wins” fails unit and evidence checks.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

All arithmetic first uses centimetres. Converting only after subtraction avoids mixing **unit**s. The record supports available, used and remaining lengths under its no-loss assumption. It does not support a promise about winning.

A numerically correct total elsewhere in a draft cannot rescue a wrong **unit** or an invented **claim**.

What this example does not establish

Keep the stated limit

Real cutting losses or inaccurate roll lengths would change this toy calculation; the assumption must be declared.

Change one thing in the pictured case

Change one thing

Change: Make ten bookmarks instead of eight.

Predicted result

Prediction: 100 cm, or 1 m, remains.

Why the result follows

Reason: 600–10×50=100 cm.

Ask while showing the picture

Ask while showing the picture · 8

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Verify Important Outputs, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: An answer says six groups of four make 28.

2. State the decision to explain

Write the question to explain: Verify **independent**ly.

3. Reveal the worked reasoning

Reveal after **prediction**s: Six times four is 24. Draw six groups or add four six times. Confident wording does not change arithmetic.

Hold back the independent task

Keep for an **independent** attempt: An assistant says the school library opens at 8, but today's official notice says 9. Which **source** should guide today's visit? Explain the check.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write **unit**s and denominators for arithmetic, trace conditions for decisions, and identify supporting **source**s for **claim**s.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before **independent** assessment.

Model these guided questions

Four rolls: supported by R. Starting 8 m: supported by 4 × 2 m. Used 600 m: contradicted; it is 600 cm = 6 m. Remaining 2 m: supported by 8 – 6, with no cutting loss. Everyone enjoys it: unresolved.

Correctness of one number does not validate another; show each calculation. Exit: loss is 12 × 2 cm = 24 cm. Remaining is 800 – 600 – 24 = 176 cm. The no-cutting-loss assumption changed.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve format versus factual support from Week 26.
7 to 20 minutes	Split the ribbon draft into five claims and complete W1.
20 to 33 minutes	Trace the evidence diagram and work through centimetres and metres.
33 to 45 minutes	Complete W2 and check used plus remaining against starting length.
Session 2 start 0 to 8 minutes	Retrieve why unresolved does not mean false.
8 to 23 minutes	Complete the guided audit and W3. Trace repeated claims to their common origin.
23 to 38 minutes	Collect W4 independently, requiring calculations, units and separate claim judgments.
38 to 45 minutes	Complete W5 and the changed-loss exit; retain the changed assumption.

Reduce help deliberately

Completion practice

Completion task: A report calculates 12 × 50 = 600 and labels the result metres. Use R to correct the **unit**, convert the result to metres and calculate the remaining length.

Give a second arithmetic route or a check using starting, used and remaining lengths.

Reasoning to listen for

Teacher check: Each piece is 50 cm, so 12 × 50 cm = 600 cm = 6 m. Starting length is 4 × 2 m = 8 m; remaining is 8 – 6 = 2 m. A check is 6 m used + 2 m remaining = 8 m starting; alternatively twelve half-metre pieces use 6 m.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Verifying **claim**s and quantities. Proposed explanation: “A **citation** proves every nearby **claim**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Inspect the actual **source** and the specific **claim** it supports.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Correct the draft and identify which part needs another **source** rather than a calculation.

Reasoned teacher answer

It should say 200 cm or 2 m remains. Winning needs genuine result **evidence**; remove the unsupported promise.

Changed-case reasoning: 100 cm, or 1 m, remains. 600–10×50=100 cm.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Verify Important Outputs investigation and use the first unsupported step to choose a teaching response.

Predict first, test honestly, and explain what the **evidence** means.

1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Use a two-**source** checklist on a generated animal fact.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the **independent** question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome

Record	What to capture
Next test	One justified change and a new prediction

Diagnose and reteach

> **Find the unsupported claim**
>
> **CLAIM** TO REPAIR A confident answer is stronger **evidence** than a relevant notice.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A **citation** proves every nearby **claim**: Inspect the actual **source** and the specific **claim** it supports.

The number is enough without a **unit**: 600 cm and 600 m differ by a factor of 100.

An unsupported **claim** must be false: Unresolved means the supplied **evidence** does not settle it.

Three copied **claim**s are three **independent** checks: Find the origin and seek an **independent** record or calculation.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Verify Important Outputs questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Six times four is 24. Draw six groups or add four six times. Confident wording does not change arithmetic.

Independent question

Independent question · withhold the key

An assistant says the school library opens at 8, but today's official notice says 9. Which

source should guide today's visit? Explain the check.

Expected independent answer

> **Teacher key · Expected independent answer**
>
> Use today's official notice, which directly addresses the current opening time. The assistant's fluent wording does not override the relevant checked **source**.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as **independent** mastery; correct unsafe or unsupported **claim**s regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this **claim**: “A confident answer is stronger **evidence** than a relevant notice.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

18/24=3/4=75%; also 0.75×24=18. Write '18 of 24 specified tests passed (75%).' Do not imply this guarantees future performance. The corrected arithmetic still depends on the test records being accurate.

Explanations for the additional workbook examples

W1 Separate source facts from derived quantities

Workbook question · W1

Using R, identify a directly stated fact and a quantity that must be calculated. Give the **source** or expression for each. Explain why a **citation** to R does not verify the **claim** that every participant will enjoy the activity.

Teacher reasoning · W1

A directly stated fact is four rolls, 2 m per roll or twelve 50 cm pieces. A derived quantity is 8 m starting length, 6 m used or 2 m remaining, supported by the relevant calculation from R.

R contains no enjoyment **evidence**, so citing it does not support everyone enjoying the activity.

W2 Repair a unit error

Workbook question · W2

A report calculates 12 × 50 = 600 and labels the result metres. Use R to correct the **unit**, convert the result to metres and calculate the remaining length.

Give a second arithmetic route or a check using starting, used and remaining lengths.

Teacher reasoning · W2

Each piece is 50 cm, so 12 × 50 cm = 600 cm = 6 m. Starting length is 4 × 2 m = 8 m; remaining is 8 − 6 = 2 m. A check is 6 m used + 2 m remaining = 8 m starting; alternatively twelve half-metre pieces use 6 m.

W3 Do not count copied claims as separate evidence

Workbook question · W3

Three posters say 600 metres were used, but all copied one draft. A fourth poster cites card Z, which is unavailable. Explain why these posters do not settle the quantity.

Name the available **evidence** and check you would use instead.

Teacher reasoning · W3

The first three are copies of one **claim**, not **independent evidence**. Z cannot be inspected, so the fourth **citation** cannot currently verify the result.

Inspect supplied R and recompute twelve 50 cm pieces as 600 cm or 6 m; verify **source**, **unit**s and the no-loss assumption.

W4 Audit a new model with mass units

Workbook question · W4

Fictional craft card M states five sealed bags of modelling material at exactly 250 g per bag. Three models each use 300 g, with no other loss.

A draft says “M supplies 1250 g; the models use 900 g; 350 kg remains; every model will win a prize.” Check all four **claim**s, show the remaining mass in grams, and explain the status of the prize **claim**. Use 1 kg = 1000 g.

Teacher reasoning · W4

1250 g is supported by 5 × 250 g; 900 g is supported by 3 × 300 g. Remaining mass is 1250 – 900 = 350 g, so 350 kg is contradicted by the **input**s and **unit** conversion: 350 kg would be 350000 g, greater than the starting mass.

The prize **claim** is unresolved because M gives no prize **evidence**; it is not established true or false by the material counts.

W5 State what a missing source prevents you from knowing

Workbook question · W5

Another draft **claim**s that the amount left in W4 accounts for all real-world waste, citing a missing instruction card. Explain which result can still be checked from M and which **claim** remains unresolved.

What information would be needed before revising the no-loss model?

Teacher reasoning · W5

The no-other-loss model in M can be checked: 350 g remains under its stated assumptions. The assertion that this includes all real-world waste is unresolved without relevant waste information.

Obtain the actual instruction or measurement record specifying additional loss, its **unit**s and what stage it covers; do not invent a waste amount or subtract it twice.

Assess independent transfer

Independent question · withhold the key

Selected case: Fictional craft card M states five sealed bags of modelling material at exactly 250 g per bag. Three models each use 300 g, with no other loss.

A draft says “M supplies 1250 g; the models use 900 g; 350 kg remains; every model will win a prize.” Check all four **claim**s, show the remaining mass in grams, and explain the status of the prize **claim**. Use 1 kg = 1000 g.

Selected case answer

Teacher key · Selected case answer

1250 g is supported by 5 × 250 g; 900 g is supported by 3 × 300 g. Remaining mass is 1250 – 900 = 350 g, so 350 kg is contradicted by the **input**s and **unit** conversion: 350 kg would be 350000 g, greater than the starting mass.

The prize **claim** is unresolved because M gives no prize **evidence**; it is not established true or false by the material counts.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or **independent**. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Verify Important Outputs to Helpful Classroom Sorter, adapt support and plan a later changed-case check.

Helpful Classroom Sorter

Require human confirmation for important sorter **output**s and attach the **evidence** record used to approve each rule or label.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct **independent** performance.

Bridge to the next week

Next: Week 28: Humans Set Boundaries. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Use a **claim**/**evidence**/decision frame and write **unit**s beside every number. Model the 100 cm to 1 m conversion before combining quantities.

Extension: explain why a plausible-looking answer can pass a size check yet still be wrong, using a different incorrect remaining length.

W4 requires 1250 g, 900 g, 350 g remaining, rejection of 350 kg with a **unit** or size explanation, and an unresolved prize **claim**.

After support change each model to use 350 g: require 1050 g used and 200 g remaining, still no **evidence** about prizes. Confirm that pupils change the affected quantities without changing unrelated **claim**s.

Week 28 distinguishes checking whether an action is sensible from checking who has authority to approve it.

Plan the delayed return

Suggested return points in this level: ai-1/28, ai-1/30, ai-1/33. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a **claim**ed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier **independent** response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Check sorter calculations with explicit **unit**s and verify explanation **claim**s against their actual catalogue **evidence**.

TEACHER GUIDE · 7 OF 9

Prepare, teach and assess

> Use this preparation summary alongside the detailed Verify Important Outputs lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Separate an answer into **claim**s and connect each **claim** to relevant **evidence**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Before class, solve the supported practice: A report calculates 12 × 50 = 600 and labels the result metres. Use R to correct the **unit**, convert the result to metres and calculate the remaining length.

Give a second arithmetic route or a check using starting, used and remaining lengths.

Supported practice answer

Reasoned practice answer: Each piece is 50 cm, so 12 × 50 cm = 600 cm = 6 m. Starting length is 4 × 2 m = 8 m; remaining is 8 − 6 = 2 m.

A check is 6 m used + 2 m remaining = 8 m starting; alternatively twelve half-metre pieces use 6 m.

Check every learner before discussion: Use the worked case for Verifying **claim**s and quantities. Proposed explanation: “A **citation** proves every nearby **claim**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Inspect the actual **source** and the specific **claim** it supports.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Make ten bookmarks instead of eight. Require a **prediction** and a reason before comparing. Expected result: 100 cm, or 1 m, remains. Reason: 600–10×50=100 cm.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

TEACHER GUIDE · 8 OF 9

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as **independent** mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved **independent** teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

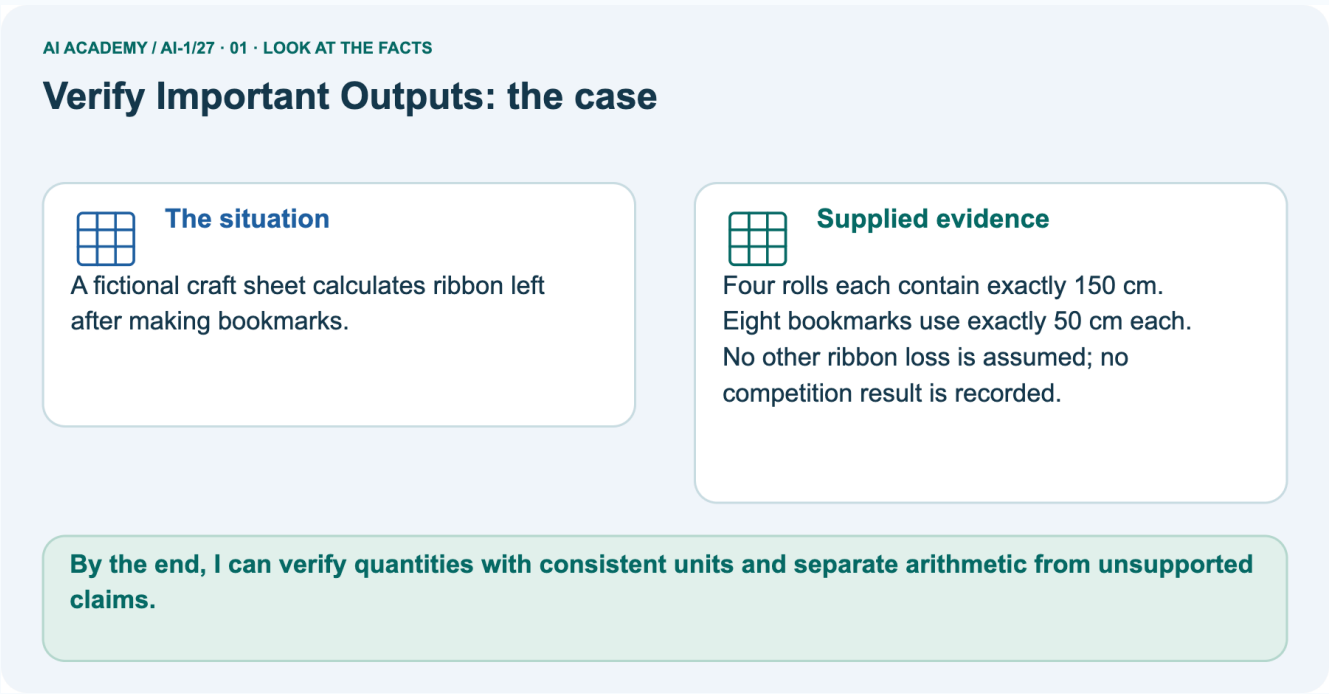
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Verify Important Outputs graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate **independent** assessment for a fresh first attempt.

01. Verify Important Outputs: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional craft sheet calculates ribbon left after making bookmarks. Supplied facts: Four rolls each contain exactly 150 cm. Eight bookmarks use exactly 50 cm each.

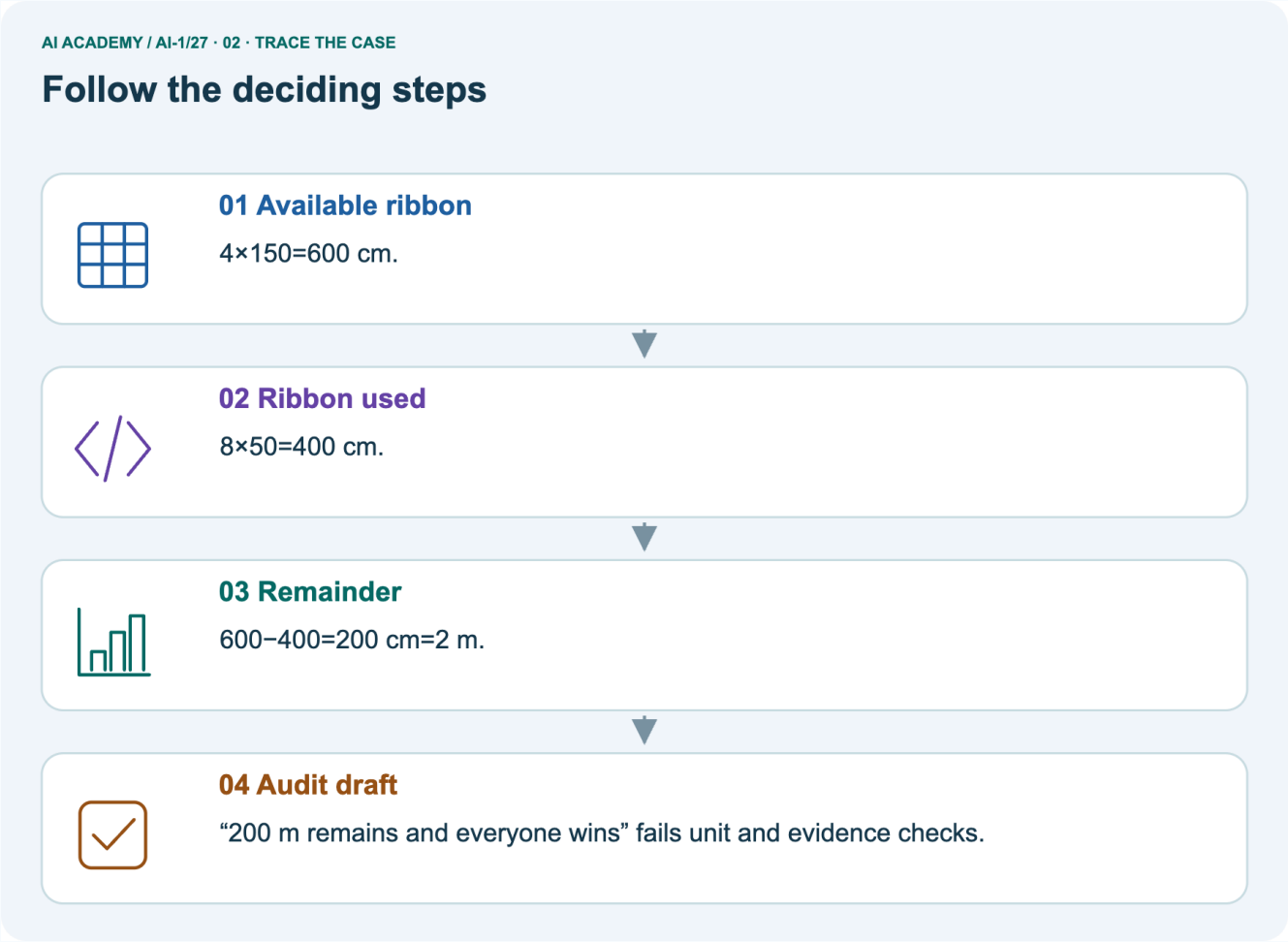
No other ribbon loss is assumed; no competition result is recorded.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Available ribbon: 4×150=600 cm. → Ribbon used: 8×50=400 cm. → Remainder: 600−400=200 cm=2 m. → Audit draft: “200 m remains and everyone wins” fails **unit** and **evidence** checks.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Ribbon quantities in one consistent unit



Follow the labelled picture with the matching explanation. Open the vector diagram

Ribbon quantities in one consistent **unit**. **Unit**s: centimetres. Available: 600. Used: 400. Remaining: 200.

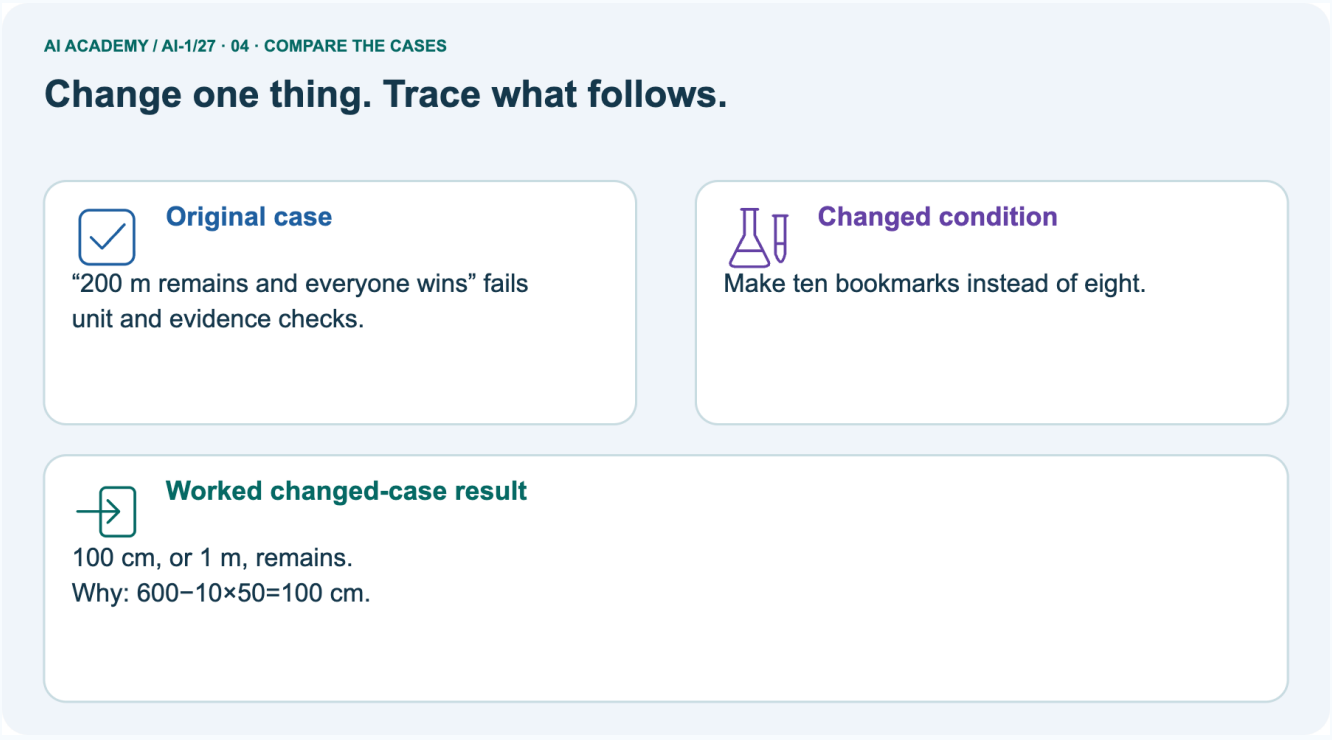
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, **unit**s and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: “200 m remains and everyone wins” fails **unit** and **evidence** checks. Changed condition: Make ten bookmarks instead of eight. Expected result: 100 cm, or 1 m, remains. Reason: 600−10×50=100 cm.

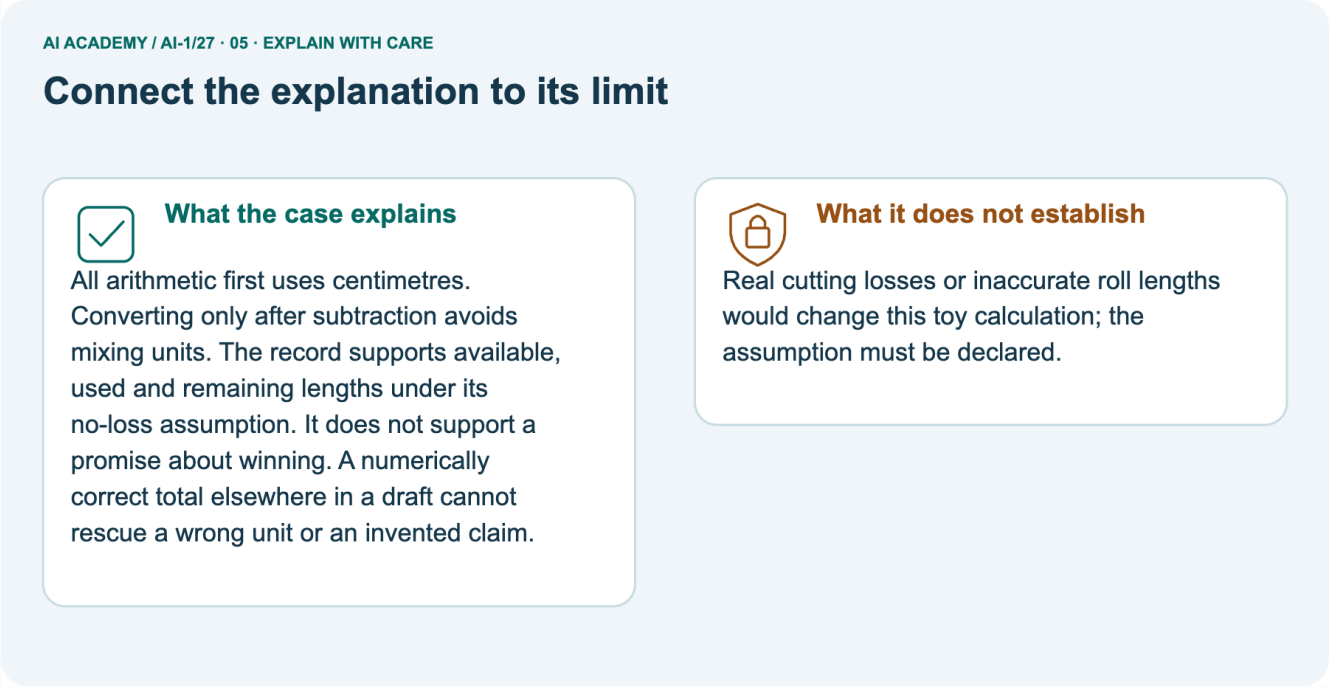
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: 600−10×50=100 cm.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

All arithmetic first uses centimetres. Converting only after subtraction avoids mixing **unit**s. The record supports available, used and remaining lengths under its no-loss assumption. It does not support a promise about winning.

A numerically correct total elsewhere in a draft cannot rescue a wrong **unit** or an invented **claim**. Limit: Real cutting losses or inaccurate roll lengths would change this toy calculation; the assumption must be declared.

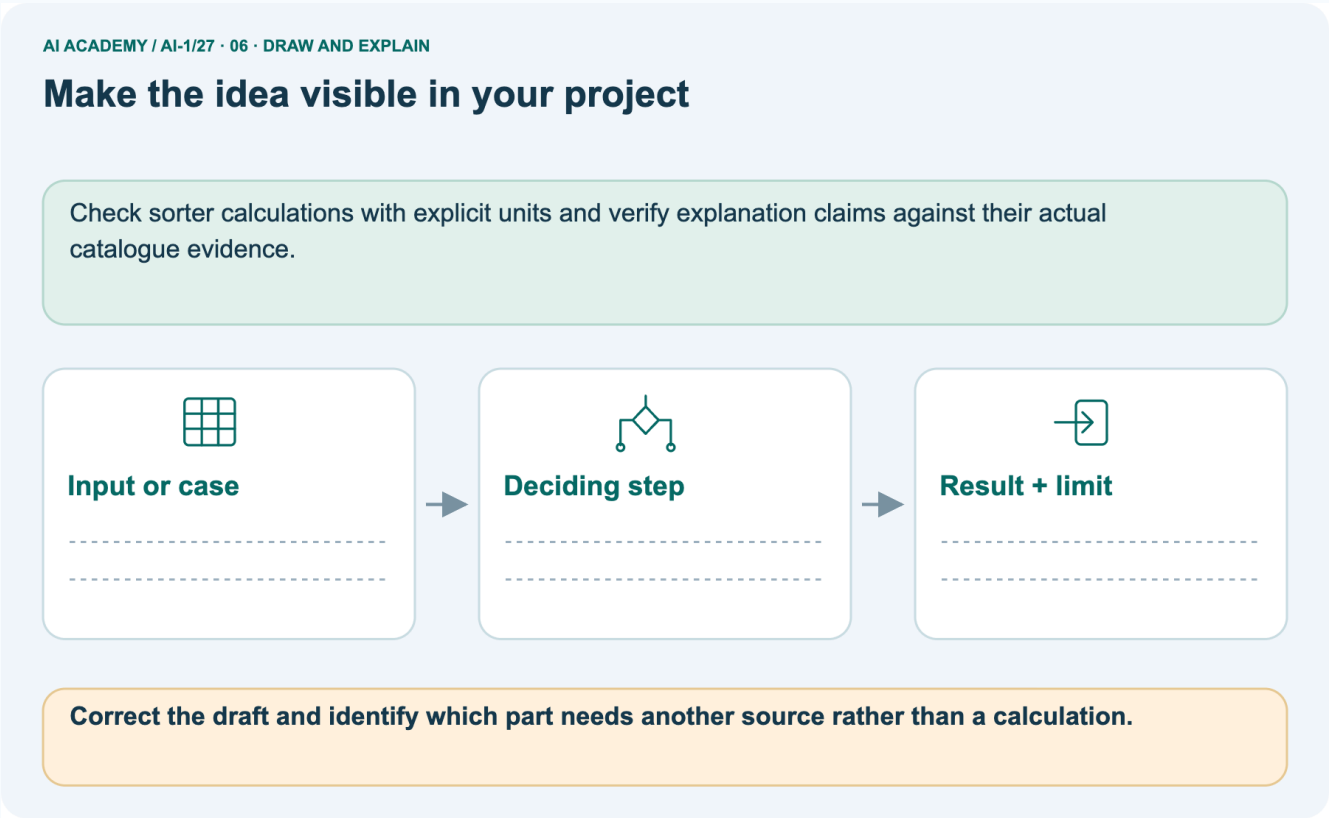
Explain the case, then point to a limit on the **claim**.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the **claim**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Real cutting losses or inaccurate roll lengths would change this toy calculation; the assumption must be declared.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Check sorter calculations with explicit **unit**s and verify explanation **claim**s against their actual catalogue **evidence**. Draw your **input**, deciding step, result and limit.

Supported question: Correct the draft and identify which part needs another **source** rather than a calculation.

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Supported practice key: It should say 200 cm or 2 m remains. Winning needs genuine result **evidence**; remove the unsupported promise.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.